

CLASS 08 / OCT 13 / U05

ATTRACTORS

2D FIELD + PLOTTER

P3 CONTINUES — A FIELD FIRST, THEN LINES A PEN CAN DRAW

P3b DUE

P3c + P3d ISSUED TODAY

AN ATTRACTOR IS A NUMBER

For every point: how far is it from something? Use that number to drive a property.

THE WHOLE IDEA

```
for each point p:  
    d = distance(p, a)  
    value = f(d)
```

Everything else this week
is a choice of f.

MEASURE IT FIRST

$$d = \sqrt{(px - ax)^2 + (py - ay)^2}$$

Pythagoras, once per point. The answer is in model units.

Index told you where in the sequence. Distance tells you how far from something outside the collection.

LAST WEEK YOU BUILT THE POSITIONS. THIS WEEK YOU MEASURE THEM.

READ THE FIELD BEFORE YOU CHANGE ANYTHING

Distance is a value at every point. Look at it as grey before it becomes geometry.

01 MEASURE

Every point on a flat 2D grid gets a number: how far to the attractor.

02 DISPLAY

Show those numbers as grey.
This is a drawing already.

03 REMAP

Push that range into the range your property needs.

04 DRIVE

Only now does anything change. The field was finished first.

THE FIELD, READ AS A GREY VALUE ACROSS THE PLANE · NEAR → FAR



IF THE GREY IMAGE IS NOT ALREADY A GOOD DRAWING, THE LINES WILL NOT SAVE IT.

P3c requires the grayscale field, with a legend, beside the line drawing. That order is the assignment.

REMAP, AND SAY WHERE YOU CLAMPED

Distance comes out in model units. Your property needs a specific range.

THE OPERATION

```
t    = (d - dmin) / (dmax - dmin)
t    = clamp(t, 0, 1)
out  = outmin + t × (outmax - outmin)

d    0 ... 8.4 units    ← input
out  2 ... 12 mm length ← output
```

SUBTRACT. DIVIDE. MULTIPLY. ADD.

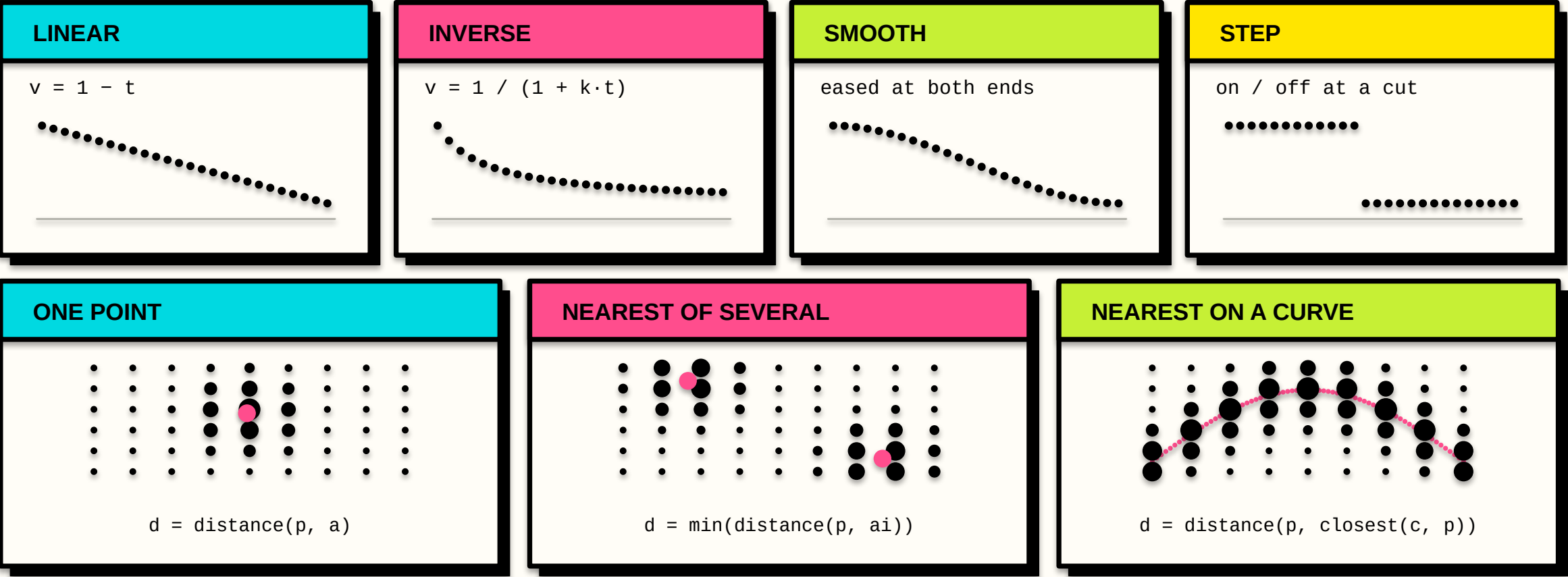
Nothing here is new. It is the arithmetic of a transformation, applied to a distance instead of a coordinate.

Clamping is a decision, not a safety net. Decide what happens past the end of your range and say so.

P3c REQUIRES INPUT AND OUTPUT RANGES, CLAMPING AND FALLOFF LABELLED ON THE SHEET.

FALLOFF, AND WHAT YOU MEASURE TO

f shapes the drawing. The attractor type changes only what the distance is measured to.



SAME GRID, SAME MAPPING, THREE ATTRACTORS: P3c SHOWS WHAT THE ATTRACTOR ALONE CHANGES.

A PLOTTER DRAWS LINES, NOT PIXELS

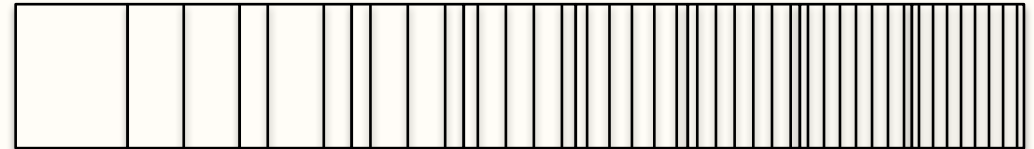
A screen can show any grey. A pen is either down or up.

ON SCREEN — A VALUE PER PIXEL



Every sample can be any grey. Nothing needs to be drawn.

ON PAPER — A STROKE OR NOTHING



Every stroke is the same darkness. Grey comes from how many lines, how long and how close.

SO THE FIELD MUST BECOME GEOMETRY: CURVES WITH A START AND AN END. AN IMAGE WILL NOT PLOT.

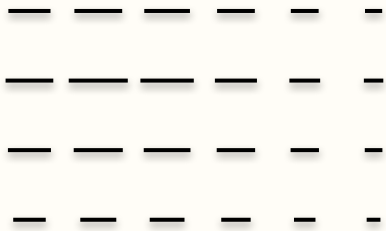
The grayscale field stays on the sheet as the legend. The line drawing is what the pen draws.

TRANSLATE THE FIELD INTO LINES

Same field, same remap. Only what the output number drives changes. Pick one.

LINE LENGTH

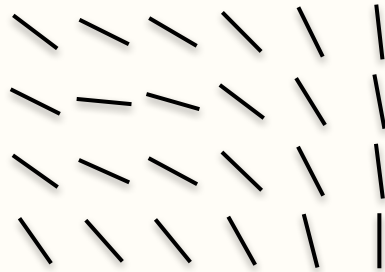
length = out



Long near, short far.

ROTATION

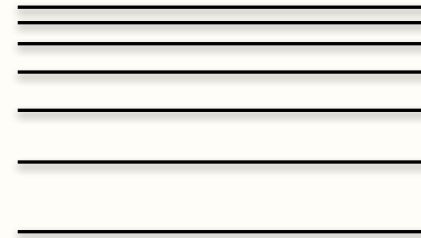
angle = out



Lines turn with distance.

SPACING

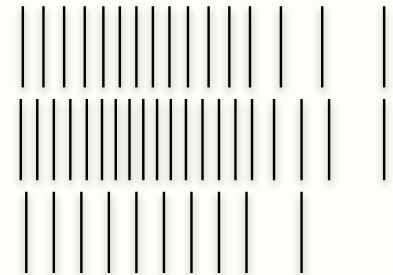
gap = out



Rows open up with distance.

HATCH DENSITY

lines per cell = out



More strokes, darker cell.

```
t = remap(distance(p, a))      # the field, unchanged
line(p, angle, length)         # the last line is the translation
```

EXPORT AT SIZE. TEST SMALL. FIX IT IN THE SVG.

The SVG is the drawing the plotter receives. Check it before the pen touches paper.

01 EXPORT THE SVG

CURVES

Keep strokes as curves. No mesh, no filled shape.

SIZE

17 × 11 in, or the confirmed plotter size, at 100%.

UNITS

Know the file unit. Measure one known line.

PEN

One pen weight, or one layer per declared pen.

MARGIN

Every stroke inside the paper, with a margin.

02 TEST PLOT SMALL FIRST

CROP

A small, dense corner of the real drawing.

PEN

Check tip and ink. A dry pen skips.

PAPER

Real paper, fixed flat, right way round.

SPEED

Slower for dense areas. Watch the first strokes.

COMPARE

Test beside screen. Change the mapping, not the pen.

03 COMMON PROBLEMS — EACH ONE IS VISIBLE IN THE SVG BEFORE IT IS ON PAPER

DUPLICATE LINES

Strokes on top of each other. Remove overlaps.

TOO-DENSE HATCHING

Lines closer than the pen blot. Space them wider.

FILLS

A fill has no stroke to follow. Turn it into hatching.

TINY STROKES

Thousands of short segments. Simplify or lower the count.

ISSUED TODAY

P3c — ATTRACTOR: 2D FIELD PLOTTER DRAWING

DUE WEEK 10 — Oct 27 / 7%

Take the field off the screen. Sample it on a flat grid, turn every value into a line, and let a pen plotter draw it. Same grid, same mapping, three attractors: the drawing shows what the attractor alone changes.

FIELD

GREY FIRST

The grayscale field, with a legend, before any line.

LINES

ONE MAPPING

Length, rotation, spacing or hatch density, driven by distance.

PLOT

SVG TO PAPER

Export at plotted size, test small, then plot the sheet.

REQUIREMENTS — P3c

Visual documentation 20% · computational understanding 20% · technical execution 15% · fabrication quality 30% · experimentation 5% · requirements and identity 10%.

01

Build a 2D attractor field on a flat grid and translate it into lines a pen plotter can draw — vary line length, rotation, spacing or hatch density with the measured distance.

02

Compare single-point, multiple-point and curve attractors on the same grid, with the same mapping, so the difference comes from the attractor alone.

03

Show the grayscale field with a legend beside the line drawing. Label input and output ranges, clamping and falloff.

04

Export an SVG at the plotted size, test a small plot, then plot the final drawing at 17 × 11 inches (or the confirmed plotter size) with one pen weight or a declared set of pens.

05

Submit the plotted drawing, the SVG, a photograph or scan of the plot and the editable graph. Move an attractor or change the mapping live at midterm.

CHECKPOINT · WEEK 9 — Oct 20 — Bring a grayscale field, the line translation and a small test plot.

ALSO ISSUED TODAY

P3d — ATTRACTOR: SITE ANALYSIS WITH MIXTLI

DUE WEEK 10 — Oct 27 / 7%

Pick a real site and analyze it with Mixtli, the Blender add-on the instructor provides. Read its output as a field — the same distance-and-remap logic as P3c — and use it to make one spatial decision. Mixtli is taught in detail next class.

SITE

RECORD IT

Source, units, scale and orientation, before any analysis.

ANALYZE

ONE COMPARISON

One analysis with a legend and units, and one comparison.

DECIDE

YOUR READING

The add-on computes. The question and the reading are yours.

REQUIREMENTS — P3d

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

Pick a site you can document and state why you chose it. Record the source of the site model or data, its units, scale and orientation.

02

Analyze the site with the Mixtli add-on for Blender, provided by the instructor. Record the analysis type, inputs and settings.

03

Make one 17 × 11 inch sheet with an annotated site plan or base view, the mapped analysis with a legend and units, and one comparison of settings, times or conditions.

04

Explain what the output measures, how it reads as a field — the same distance-and-remap logic as P3c — and how it informs one spatial decision on the site.

05

Submit the working file and dependencies. The add-on does the computation; the question, comparison and interpretation are yours.

CHECKPOINT · WEEK 9 — Oct 20 — Bring your chosen site, one Mixtli analysis with a legend and a stated analysis question.

NOW — FIELD TO PAPER

Rest of the class: build the field, translate it, export it, test it.

✓

Build the 2D field: a flat grid, distance at every sample, shown as grey with a legend

✓

Write down input and output ranges, clamp and falloff — actual numbers

✓

Translate the field into lines: length, rotation, spacing or hatch density

✓

Export an SVG at plotted size and measure one known line

✓

Run a small test plot and compare it with the screen

MOVE THE ATTRACTOR. IF YOU CANNOT PREDICT WHAT HAPPENS, STOP AND READ THE FIELD.

BEFORE NEXT CLASS

NEXT

Week 9 — Oct 20: site analysis with Mixtli, the Blender add-on the instructor provides.

P3c

Checkpoint next class. Bring a grayscale field, the line translation and a small test plot.

P3d

Checkpoint next class. Bring your chosen site, one Mixtli analysis with a legend and a stated analysis question.

MIDTERM

Week 10 — Oct 27: midterm review of Projects 1–3. Due then: GSM, P2b, P3c, P3d and MID. Do not leave prints or plots to the last week.